

120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 11: Portfolio Optimization and Risk Management using ML

Day 76: Portfolio Management with Machine Learning

1 Introduction

Portfolio management involves allocating capital across different assets in order to maximize returns while controlling risk. Traditional portfolio management relies on statistical models such as Modern Portfolio Theory and mean-variance optimization.

However, financial markets are highly complex and dynamic. Relationships between assets change over time, and new information continuously affects asset prices.

Machine learning provides powerful tools that can improve portfolio management by identifying patterns in large datasets and adapting to changing market conditions.

2 Key Components of Portfolio Management

Portfolio management typically involves three main steps:

1. Return prediction
2. Risk estimation
3. Portfolio allocation

Machine learning models can contribute to each of these stages.

3 Return Prediction with Machine Learning

Machine learning models can be trained to predict future asset returns using historical market data and engineered features.

Suppose we have a set of features describing market conditions:

$$X_t = (x_1, x_2, \dots, x_k)$$

A machine learning model learns a mapping

$$\hat{r}_{t+1} = f(X_t)$$

where $\hat{r}_{t+1}$ represents the predicted return for the next time period.

Common models used include:

- Linear regression with regularization
- Random forests
- Gradient boosting
- Neural networks

4 Risk Forecasting

Risk forecasting is another key component of portfolio management. Machine learning models can predict volatility or downside risk using historical data.

For example, predicted volatility can be written as

$$\hat{\sigma}_{t+1} = g(X_t)$$

where $g(\cdot)$ represents a machine learning model.

These predictions can be used to adjust portfolio exposure dynamically.

5 Portfolio Allocation

Once expected returns and risk estimates are obtained, portfolio weights can be determined.

A simple allocation rule may be based on predicted returns:

$$w_i = \frac{\hat{r}_i}{\sum_{j=1}^n |\hat{r}_j|}$$

This assigns higher weights to assets with stronger predicted returns.

Another approach combines expected returns and risk:

$$w_i = \frac{\hat{r}_i}{\hat{\sigma}_i}$$

which allocates more capital to assets with high expected return and lower predicted risk.

6 Dynamic Portfolio Rebalancing

Machine learning enables dynamic portfolio rebalancing by updating predictions as new data arrives.

At each time step:

1. Update market features
2. Predict returns and risk
3. Recompute portfolio weights
4. Rebalance the portfolio

This creates an adaptive investment strategy that responds to changing market conditions.

7 Advantages of ML-Based Portfolio Management

Machine learning offers several advantages:

- Ability to capture nonlinear relationships
- Use of large and diverse datasets
- Adaptive learning from new information
- Improved prediction accuracy

These features make ML a valuable tool for modern quantitative portfolio management.

8 Conclusion

Machine learning is transforming portfolio management by enabling more data-driven and adaptive investment strategies.

By combining return prediction, risk estimation, and dynamic allocation, ML-based portfolio management can improve decision-making in complex financial markets.